

Hiya Dullu



8076778508



hiyadullu05@gmail.com



LinkedIn



github

Education

Vellore Institute Of Technology, Bhopal

2023-2027

Bachelor of Technology in Computer Science and Engineering (GPA: 8.51)

Technical Skills

Languages: Python, Java, JavaScript, SQL

Technologies: : FastAPI, PySpark, Scikit-Learn,, TensorFlow, OpenCV, PostgreSQL, MySQL, SQLite, Docker, AWS (EC2, S3, IAM), CI/CD (GitHub Actions), Git, Ubuntu

Concepts: : Machine Learning, Deep Learning, Computer Vision, Large Language Models (LLM), RESTful APIs, Software Deployment

Experience

Thales Group, Noida

May 2026 – Present

SDE Intern – HSM Payment Department

- Designed an unsupervised anomaly detection pipeline for payment fraud detection across transaction feature embeddings, improving fraud recall to 85% in internal benchmarking.
- Integrated a local Mistral LLM via Ollama REST APIs to generate interpretable forensic summaries for anomalous transactions, reducing manual investigation effort.
- Developed an interactive dashboard featuring real-time risk KPIs, live variance distribution scatterplots, and automated audit logging.
- Contributed to modular backend development and low-latency inference workflows in collaboration with cross-functional teams.

Projects

Emotion Recognition using Swin Transformer | Python, PyTorch, OpenCV, Transformers

- Built a two-stage emotion recognition pipeline using a Swin Transformer trained on FER-2013 (74% accuracy) and extended to video-based classification on IEMOCAP and RAVDESS via transfer learning.
- Implemented temporal pooling over 8-frame sequences to enable efficient video-level emotion prediction, avoiding complex sequence models while maintaining strong performance.
- Developed an end-to-end pipeline (preprocessing, frame extraction, label mapping, evaluation) and identified data leakage in validation (90% accuracy), proposing video-level splitting to ensure robust generalization.

QR-Enabled Digital POS and Ledger Management System | Node.js, Express, PostgreSQL, EJS, Passport.js, Render, Gemini API

- Architected a modular, web-based restaurant management platform integrating POS systems, QR-based customer ordering, admin dashboards, and automated onboarding workflows.
- Built scalable REST APIs using Node.js and Express with PostgreSQL for transactional data, authentication via Passport.js, and dynamic QR-code-driven customer menus, reducing manual order handling and improving operational efficiency
- Implemented automated deployment notifications and admin-controlled onboarding pipelines, enabling faster restaurant go-live times, with planned AI-driven insights using Gemini API for customer behavior and market trend analysis.

IEEE Conference Paper – Cardiovascular Disease Prognosis Using Ensemble Machine Learning

- Developed a machine learning–based clinical decision support system using the UCI Cleveland Heart Disease dataset (14+ clinical features)
- Implemented Logistic Regression, SVM, Random Forest, XGBoost, and Neural Networks for comparative analysis. Designed a soft-voting ensemble model achieving 92% accuracy and 0.91 macro F1-score.
- Published findings in an IEEE-affiliated international conference, demonstrating real-world healthcare analytics application.

Leadership & Extracurricular Experience

Software Development Club – General Secretary

- Managed end-to-end logistics and operations for technical events and workshops, coordinating cross-functional teams.

- Developed backend components for the club website, supporting content delivery and internal coordination.

Android Club – R&D Department Lead

- Led technical content strategy for workshops, curating structured materials aligned with Android and backend development practices.

Certifications

- | | |
|--|----------|
| • Python Pro Bootcamp – Udemy | May 2024 |
| • AWS Certified AI Practitioner – Ethnus | May 2025 |
| • IBM Generative AI – IBM | May 2025 |
| • The Bits and Bytes of Computer Networking – Coursera | Nov 2025 |